

Open Reduction and Internal Fixation (ORIF)

1. Indications for ORIF

- **Displaced unfavourable fractures:** Fractures where broken bone segments do not align naturally.
- **Severely atrophic edentulous mandibles:** Extremely resorbed jawbones without teeth.
- **Complex facial fractures:** Complicated breaks involving multiple areas of the face.
- **Condylar fractures:** Breaks in the mandibular condyle, the hinge-like structure of the jaw.
- **Non-unions or malunions:** Healing failure or improper alignment after a fracture.

2. Approaches for ORIF

- **Wire osteosynthesis:** Fixation using wires to stabilize fractured bones.
- **Intraoral approach:** Accessing the fracture through the inside of the mouth.
- **Submandibular approach:** Surgical access below the lower jaw.
- **Retromandibular approach:** Approach through the area behind the lower jaw.
- **Preauricular approach:** Access near the front of the ear.

Closed Reduction (CR)

1. Indications for CR

- **Nondisplaced favourable fractures:** Fractures where the bone segments remain in alignment.
- **Grossly comminuted fractures:** Highly fragmented fractures, requiring minimal bone stripping for healing.
- **Fractures in children:** Ensures protection of developing teeth and tooth buds.
- **Coronoid fractures:** Breaks in the coronoid process of the mandible; treated only if it impinges on the zygomatic arch.

2. Procedures for CR

- **Erich arch bars:** Devices fixed to teeth for fracture stabilization.
- **Bridle wires:** Wires used to link fractured segments.
- **Ivy loops:** Wire loops used for jaw fixation.
- **Biphasic pin fixation:** Pins inserted to immobilize fractured bones.

Hypothyroidism

1. Causes of Hypothyroidism

- **Primary Hypothyroidism:** The problem originates in the thyroid gland.
 - **Autoimmune (spontaneous atrophic):** Body's immune system attacks the thyroid, leading to shrinking and loss of function.

- **Iodine deficiency:** Lack of iodine in the diet, essential for thyroid hormone production.
- **Post-131I therapy:** Hypothyroidism following radioactive iodine treatment for hyperthyroidism.
- **Hashimoto's thyroiditis:** Chronic autoimmune inflammation of the thyroid gland.
- **Genetic defects (dyshormonogenesis):** Inherited errors in hormone production.
- **Drugs (lithium, amiodarone):** Medications that interfere with thyroid function.
- **Secondary Hypothyroidism:** The issue arises from the pituitary or hypothalamus.
 - **Pituitary gland destruction:** Damage to the pituitary due to surgery or tumors.
 - **Hypothalamic disorders:** Conditions affecting the hypothalamus, which regulates the pituitary.

2. Clinical Features

- **General:**
 - **Puffy face:** Swollen appearance due to fluid retention.
 - **Goiter:** Enlarged thyroid gland.
- **GI (Gastrointestinal):**
 - **Ascites:** Fluid buildup in the abdomen.
- **Cardio (Cardiovascular):**
 - **Bradycardia:** Slow heart rate.
- **Neuromuscular:**
 - **Delayed reflexes:** Slower than normal response to reflex tests.
- **Skin:**
 - **Alopecia:** Hair loss.
 - **Myxedema:** Swelling due to the accumulation of mucopolysaccharides in the skin.
- **Reproductive:**
 - **Menorrhagia:** Heavy menstrual bleeding.
- **Hematology:**
 - **Macrocytosis:** Larger than normal red blood cells.

3. Investigations

- **TSH (Thyroid-Stimulating Hormone):** High in primary hypothyroidism (feedback to a non-functional thyroid), low in secondary hypothyroidism (due to pituitary dysfunction).
- **TPO (Thyroid Peroxidase):** Enzyme targeted by antibodies in autoimmune hypothyroidism.
- **Hyponatremia:** Low sodium levels in the blood.

Cleft Lip and Palate

1. Aetiology

- **Environmental factors:**
 - **Antiepileptic drugs:** Medications for controlling seizures, like phenytoin and diazepam.
 - **Rubella:** A contagious viral infection that can harm a developing fetus.
 - **Retinoic acid:** A compound derived from vitamin A that can affect fetal development.
- **Genetic factors:**
 - **First-degree relatives:** Immediate family members like parents or siblings.
 - **Pierre-Robin sequence:** A condition with a small jaw, tongue displacement, and breathing issues.
 - **Stickler syndrome:** A genetic disorder affecting connective tissues.
 - **Apert syndrome:** A condition with fused skull bones and limb abnormalities.
 - **Treacher Collins syndrome:** A disorder affecting facial structure development.

2. Pathophysiology

- **Mesenchymal migration:** Movement of embryonic connective tissue to form structures.
- **Medial nasal process:** Central part of the nose during embryonic development.
- **Maxillary process:** Part of the upper jaw forming the face and palate.
- **Palatal shelf:** Horizontal structures that develop into the roof of the mouth.

3. Clinical Features

- **Visible fissure:** A noticeable gap or split.
- **Misaligned teeth:** Teeth that are not in proper position.
- **Nasal distortion:** Abnormal shape or position of the nose.
- **Recurrent ear infections:** Frequent infections in the middle ear.

4. Management

- **Cleft Lip:**
 - **Millard rotation-advancement flap:** A surgical technique that rotates and advances tissues for repair.
 - **Tennison-Randall repair:** A method using a triangular flap to lengthen the lip.
- **Cleft Palate:**
 - **Double-opposing Z-plasty:** A zigzag incision to realign muscles and improve function.
 - **Vomer flap:** Tissue from the vomer bone (part of the nasal septum) used to close the hard palate.

Carotid Body Tumors (CBTS)

1. Aetiology

- **Chronic hypoxic stimulation:** Prolonged low oxygen levels in the body, often due to respiratory or cardiovascular issues.
- **Genetic predisposition:** An inherited tendency to develop certain conditions, often seen with a family history of similar tumors.

2. Clinical Features

- **Age of onset:** The most common age for development of these tumors is around 45 years.
- **Asymptomatic mass:** A lump or swelling in the anterior neck (front part of the neck) that does not cause symptoms in the early stages.
- **Fontaine sign:** A sign indicating the mass is fixed to the carotid bifurcation, meaning it is not freely movable.
- **Bruit:** An abnormal sound, typically heard through a stethoscope, caused by turbulent blood flow through the tumor.
- **Cranial nerve palsy:** Weakness or paralysis of nerves in the head and neck, leading to symptoms like tongue paralysis, difficulty swallowing, or hoarseness.
 - **Hypoglossal nerve:** Nerve controlling tongue movements, damage causes paralysis of the tongue.
 - **Glossopharyngeal nerve:** Nerve responsible for taste and swallowing, damage affects both functions.
 - **Recurrent laryngeal nerve:** Nerve controlling the vocal cords, damage causes hoarseness.
 - **Spinal accessory nerve:** Nerve that controls the shoulder and neck muscles, damage causes shoulder drop.
 - **Sympathetic chain involvement:** Damage to the sympathetic nerves causing **Horner syndrome**, which includes ptosis (drooping eyelid), miosis (constricted pupil), and anhidrosis (lack of sweating).

3. Symptoms Associated with CBTS

- **Pain:** Discomfort that may arise as the tumor grows.
- **Hoarseness:** A raspy or abnormal voice, often due to nerve involvement affecting the vocal cords.
- **Dysphagia:** Difficulty in swallowing, which can occur if the tumor presses on structures in the neck.
- **Horner syndrome:** A combination of signs, including drooping eyelid (ptosis), constricted pupil (miosis), and absence of sweating (anhidrosis), due to sympathetic nerve damage.
- **Shoulder drop:** A lowered position of the shoulder, caused by damage to the spinal accessory nerve.

Signs and Symptoms of Shock

1. CNS Symptoms

- **Cerebral perfusion:** Blood flow to the brain.

- **Hypoxia:** Lack of oxygen in the tissues, including the brain, leading to altered mental status.
- 2. **Cardiovascular Symptoms**
 - **Hypotension:** Low blood pressure, indicating poor circulatory volume.
 - **Tachycardia:** Increased heart rate.
 - **Weak, thready pulse:** A pulse that is faint and difficult to feel, indicating reduced blood flow.
- 3. **Skin Changes**
 - **Vasoconstriction:** Narrowing of blood vessels, reducing blood flow to the skin.
 - **Mottled extremities:** Skin appearing blotchy or discolored due to poor blood flow.
 - **Pallor:** Pale skin due to reduced blood circulation.
- 4. **Respiratory Symptoms**
 - **Tachypnea:** Rapid, shallow breathing, often due to acidosis (excess acid in the body).
- 5. **Other Symptoms**
 - **Hypothermia:** Lowered body temperature, often due to reduced blood flow and sweat evaporation.
 - **Fatigue:** Tiredness due to inadequate oxygen supply to the muscles and organs.
 - **Fainting:** Loss of consciousness due to insufficient blood supply to the brain.
 - **Oliguria/anuria:** Reduced or no urine output, indicating poor kidney perfusion and vasoconstriction of afferent arterioles (small arteries leading to the kidneys).

Drugs in Management of Shock

1. **Sedatives**
 - **Relieve pain and anxiety:** Medications used to reduce pain and calm a patient.
 - **Morphine:** A strong pain-relieving medication, preferred in intravenous (IV) form during shock because it acts faster than intramuscular (IM) injections.
 - **Pethidine:** Another pain-relieving drug, effective but less commonly used than morphine.
2. **Chronotropic Agents**
 - **Low heart rate:** When the heart rate is abnormally slow.
 - **Atropine:** A medication that increases heart rate by blocking certain nerve impulses.
 - **Isoproterenol:** A drug used to stimulate the heart, increasing heart rate and improving blood circulation.
3. **Inotropic Agents**
 - **Enhance cardiac muscle contraction:** Drugs that increase the strength of heart contractions, improving blood circulation.

- **Dopamine:** A drug that can increase heart rate and blood pressure, boosting cardiac output.
 - **Dobutamine:** Another inotropic drug that improves heart function by strengthening heart muscle contractions.
 - **Benefits:**
 - **↑ Myocardial activity:** Increased activity of the heart muscle.
 - **↑ Renal blood flow via vasodilation:** Increased blood flow to the kidneys through the widening of blood vessels.
 - **Use carefully in small doses:** Caution is advised, as overuse can have harmful effects.
4. **Vasodilators**
- **Septic, traumatic, cardiogenic shock:** Types of shock where blood vessels are dilated to improve circulation and reduce pressure.
 - **Nitroprusside:** A vasodilator used to relax blood vessels and reduce blood pressure.
 - **Nitroglycerin:** A short-acting vasodilator that works similarly to nitroprusside, used to reduce heart strain.
5. **Vasoconstrictors**
- **Neurogenic shock:** Shock caused by disruption to the nervous system, often leading to low blood pressure.
 - **BP:** Blood pressure.
 - **Coronary circulation perfusion:** Blood flow to the heart muscle.
 - **Phenylephrine:** A vasoconstrictor that tightens blood vessels, increasing blood pressure.
 - **Metaramino:** A vasoconstrictor used similarly to phenylephrine.
 - **Avoid in hypovolemic and traumatic shock:** Should not be used in shock caused by blood or fluid loss, as it can worsen low blood volume.
6. **Beta-Blockers**
- **Stiff myocardium:** A condition where the heart muscle becomes less flexible, often seen in cardiogenic shock.
 - **Rapid heart rate:** Abnormally fast heart rate, often a result of shock.
 - **Propranolol:** A beta-blocker that slows the heart rate and reduces heart strain, used in cardiogenic shock.
7. **Diuretics**
- **Cardiogenic shock:** Shock due to heart failure, often leading to fluid buildup in the body.
 - **Vascular volume and filling pressure:** The amount of blood in the vessels and the pressure that helps the heart fill with blood.
 - **Avoid in hypovolemic and septic shock:** Diuretics should not be used in shock due to blood or fluid loss, as they can further reduce blood volume and worsen the condition.

Clinical Features of Tetanus

1. Incubation

- **Incubation:** The period between exposure to the bacteria and the appearance of symptoms.

2. Symptoms

- **Trismus:** Also known as "lockjaw," it is the inability to open the mouth due to muscle spasms.
- **Stiffness:** Muscle rigidity, often beginning in the jaw and neck and spreading to other parts of the body.
- **Spasms:** Sudden, involuntary muscle contractions, which can be triggered by stimuli like noise or light.
- **Apnoea:** The cessation of breathing, which can occur due to spasms affecting the diaphragm.
- **Fractures:** Broken bones, which can result from violent muscle spasms.
- **Asphyxia:** Lack of oxygen, often due to inability to breathe properly.
- **Risus sardonicus:** A "sardonic grin," or abnormal facial expression resembling a smile, often caused by spasm of the facial muscles.
- **Opisthotonys:** A severe form of muscle spasm, where the back arches and the head and heels may touch the floor.
- **Diaphragm spasms:** Involuntary contractions of the diaphragm, leading to breathing difficulties.
- **Autonomic dysfunction:** Disruption in the automatic control of bodily functions, such as heart rate, blood pressure, and breathing.

3. Diagnostic Test

- **Spatula test:** A test where a spatula is placed in the patient's throat, and a spasm occurs if the patient bites down on it due to the reflex.

4. Types

- **Generalized:** The most common form, starting with trismus and spreading to other parts of the body.
- **Neonatal:** Tetanus that occurs in newborns, usually due to infection from an unclean umbilical stump.
- **Local:** Spasms limited to the area near the injury site.
- **Cephalic:** Tetanus that primarily affects the cranial nerves, often affecting the face and head.

Actiopathogenesis of Solitary Thyroid Nodule

1. **Follicular adenoma:** A benign tumor of the thyroid gland made up of normal thyroid tissue, often presenting as a solitary nodule.
2. **Thyroiditis:** Inflammation of the thyroid gland, which can cause swelling and nodules to form.
3. **Thyroid cyst:** A fluid-filled sac within the thyroid gland, which may appear as a nodule.
4. **Colloid nodules:** Benign growths made up of thyroid cells and colloid, a gel-like substance produced by thyroid follicles.

5. **Carcinoma:** Cancerous growth in the thyroid gland, which may present as a solitary nodule.

Clinical Features

1. **Common in females (20-40 years):** Solitary thyroid nodules are more frequently seen in women, especially in the age group of 20-40 years.
2. **Euthyroid:** A state where thyroid function is normal, and there are no signs of hyperthyroidism or hypothyroidism in most cases.
3. **Symptoms:**
 - **Long-standing neck swelling:** A persistent lump or swelling in the neck that may gradually enlarge.
 - **Dyspnoea:** Difficulty breathing, which may occur if the nodule compresses the airway.
 - **Dysphagia:** Difficulty swallowing, often due to pressure exerted by the nodule on the esophagus.
 - **Obstructive signs:**
 - **Vein engorgement:** Swelling of veins, particularly in the neck, due to obstruction of blood flow.
 - **Stridor:** A high-pitched sound when breathing, caused by airway obstruction.
 - **Tracheal deviation:** Shifting of the trachea (windpipe) to one side due to pressure from the nodule.
4. **Single nodule present:** The presence of only one nodule in the thyroid gland.
 - **Hard area (Calcification):** A nodule with calcified regions that feel firm or hard to the touch.
 - **Soft area (Necrosis):** A nodule with soft regions, which may indicate tissue death or necrosis.
 - **Sudden increase in size (Possible hemorrhage):** Rapid enlargement of the nodule, which may be due to bleeding inside the nodule.
5. **Higher malignancy risk than multi-nodular goitre:** A solitary thyroid nodule has a higher likelihood of being cancerous compared to a multi-nodular goitre, which is typically benign.

Management of Shock

1. **Resuscitation:** The process of restoring the body's fluid balance and oxygenation, often through fluids or medications.
2. **Control of hemorrhage:** Stopping or controlling excessive bleeding.
3. **Extracellular fluid replacement:** Administering fluids to restore the fluids outside the cells, in the blood vessels and tissues.
4. **Correct acid-base balance:** Balancing the body's pH levels, correcting any acidosis or alkalosis.

5. **Drugs:** Medications used to manage shock, which may include vasopressors, inotropes, or other drugs to stabilize vital signs.

Treatment of Septic Shock

1. **Infection Control:** Measures taken to reduce or prevent further infection, such as antibiotics or sterile techniques.
2. **Fluid replacement:** Administering fluids to compensate for lost blood volume or fluid in the body.
3. **Vasopressor therapy:** The use of medications that constrict blood vessels to raise blood pressure in cases of shock.
4. **Vasoactive drugs:** Medications that affect blood vessel tone and heart function to help regulate blood pressure.
5. **Mechanical ventilation:** The use of a machine to assist or control breathing in patients with respiratory failure.
6. **Steroid therapy:** The use of corticosteroids to reduce inflammation and stabilize blood pressure.

Clostridium tetani and Tetanus Management

- **Clostridium tetani:** A type of anaerobic (oxygen-free) gram-positive bacterium that causes tetanus.
- **Diagnosis:** Identifying tetanus through symptoms and clinical evaluation.
- **Wound care:** Cleaning and treating any wounds to prevent infection, including tetanus.
- **Antibiotics:** Medications used to treat infections, such as **metronidazole**, often given for 10 days to kill the bacteria.
- **Passive immunization:** Providing tetanus immunoglobulin (antibodies) to neutralize toxins in the body.
- **Supportive care:** Care aimed at providing relief, such as using **diazepam** to manage muscle spasms caused by tetanus.

Universal Precautions

1. **Precautions:** Steps taken to prevent the spread of infection, particularly in healthcare settings.
2. **CGDP:** Stands for specific protocols or guidelines for infection control (e.g., **Hand Hygiene, Waste Disposal**).
3. **HR-W:** Likely refers to a specific workflow or handling process in patient management.
4. **OPD:** Outpatient Department, where patients receive non-emergency care.
5. **Disposable ins:** Refers to disposable instruments used in patient care.
6. **Reusable ins:** Instruments that are cleaned and sterilized for multiple uses.

7. **Waste disposal:** The proper disposal of medical waste to prevent contamination and infection.
8. **Hand hygiene:** The practice of thoroughly washing hands to prevent infection transmission.

Precautions in Operating Room

1. **Environmental cleaning:** The process of disinfecting and cleaning the operating room to maintain a sterile environment and reduce the risk of infection.
2. **Dental chair:** The chair used in dental procedures that must be cleaned and disinfected before and after each patient.
3. **Surgical tech:** Refers to the surgical technician or technologist, who assists during surgery by handling instruments and maintaining a sterile environment.
4. **Personnel:** Refers to the healthcare workers (doctors, nurses, technicians) involved in the surgical process, who must follow strict hygiene protocols.

Post-Exposure Prophylaxis for Health Workers

1. **Immediate action:** Quick steps taken immediately after a potential exposure to infectious agents (e.g., bloodborne pathogens) to prevent transmission.
2. **Prophylactic medication:** Medications given to prevent infection after potential exposure. Examples include **ART (Antiretroviral Therapy)** and **AZT (zidovudine)**, a drug used to prevent HIV transmission.
3. **Follow-up:** Ongoing monitoring and testing after exposure to ensure no infection has occurred.
4. **Vaccination:** Administering vaccines (such as hepatitis B) to prevent infections after exposure.

Precautions for Airborne Diseases

1. **Wear N-95 masks:** N-95 respirators are special masks designed to filter out airborne particles, including viruses, providing protection from airborne diseases.
2. **Precautions for COVID-19:** Special measures to prevent the spread of the SARS-CoV-2 virus, which causes COVID-19, including wearing masks, social distancing, and other hygiene measures.

Indications for ORIF in Mandibular Angle Fracture

1. **Displaced unfavorable fracture:** A fracture of the mandibular angle (part of the jaw) that is misaligned and needs surgical intervention to align the bone properly.

Management of Open Fractures

1. **Wound care:** Proper treatment and cleaning of the wound to prevent infection and promote healing.
2. **Antibiotics:** Medications used to treat or prevent bacterial infections, which are commonly prescribed after open fractures to prevent infection.
3. **Tetanus prophylaxis:** Administration of tetanus immunoglobulin and/or tetanus vaccine to prevent tetanus infection following an open fracture.
4. **Reduction & fixation:** Surgical procedures to realign the fractured bone and stabilize it with hardware such as pins, screws, or plates.
5. **Immobilization:** Keeping the fractured area still (using casts, splints, or other devices) to allow the bone to heal properly.
6. **Rehabilitation:** Therapy and exercises aimed at restoring function and strength to the injured area after healing.